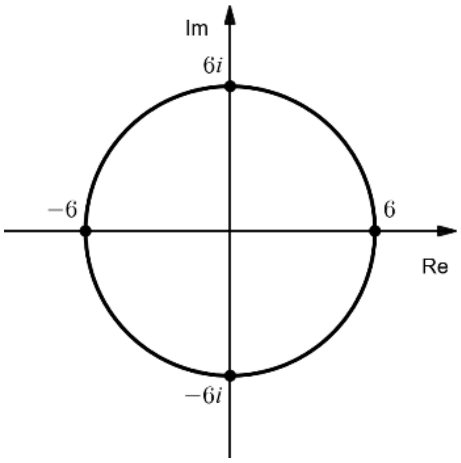
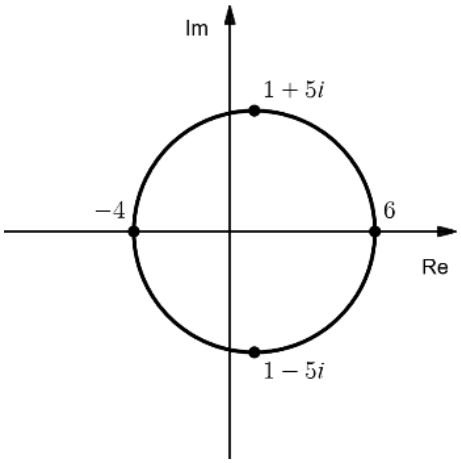
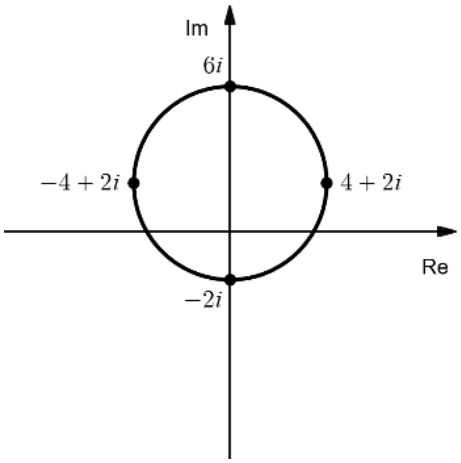
1) a)



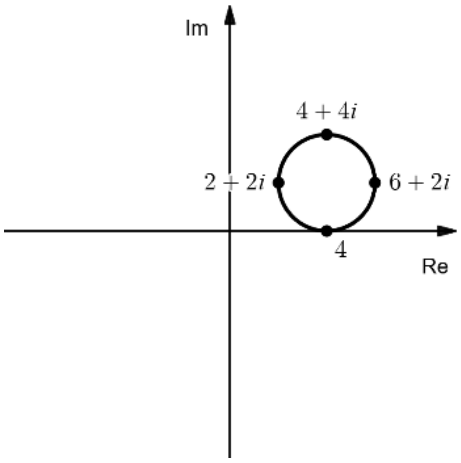
1) b)



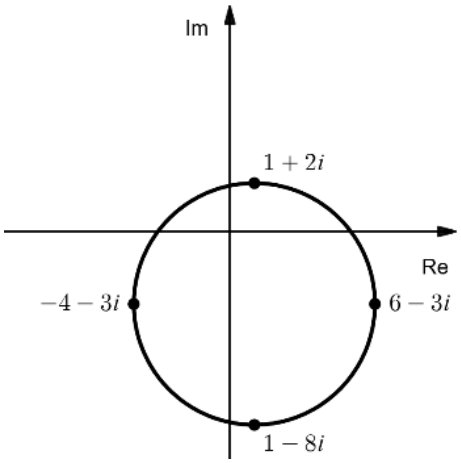
1) c)



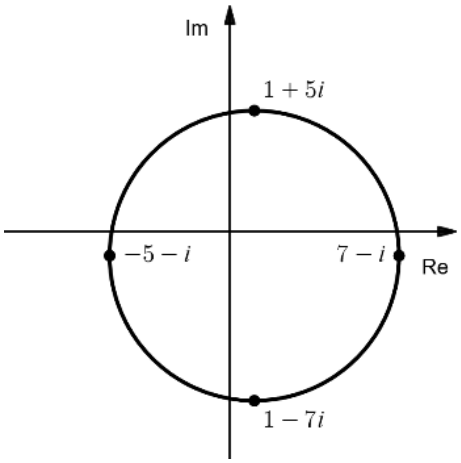
1) d)



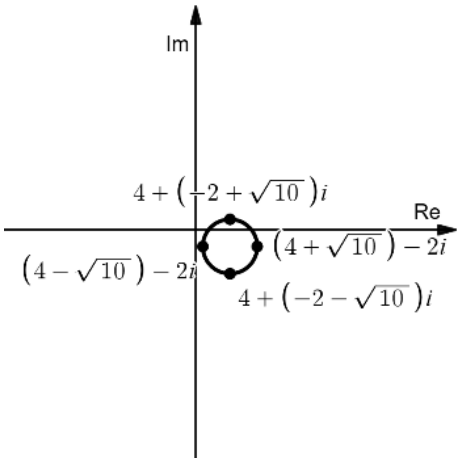
1) e)



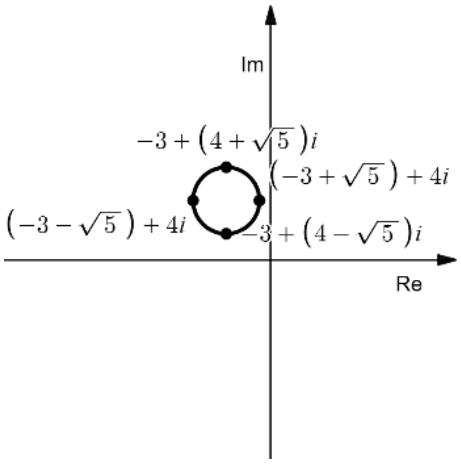
1) f)



1) g)



1) h)



2) a)

$|z| = 4$

2) b)

$|z - 3\mathrm{i}| = 5$

2) c)

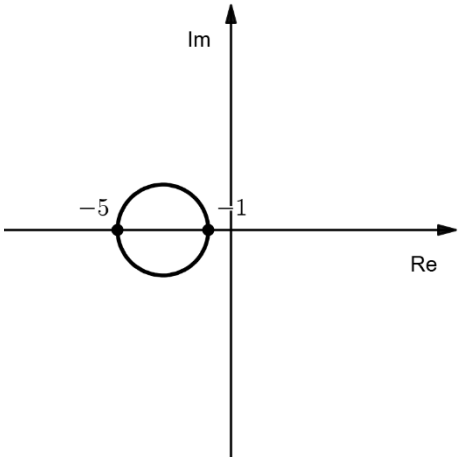
$|z - 2 + 5\mathrm{i}| = 2$

2) d)

$|z + 3 - 2\mathrm{i}| = 4$

3) a)

Centre is $(-3, 0)$ and radius is 2, so $(x+3)^2 + y^2 = 4$

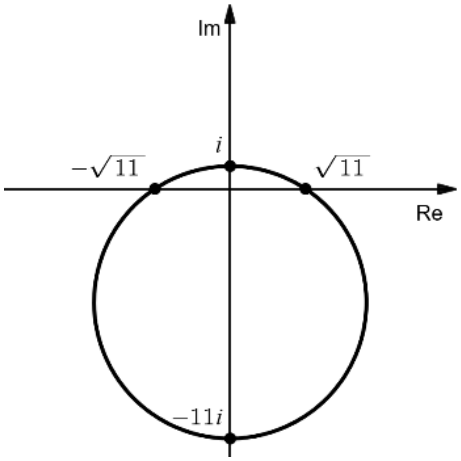


3) b)

Centre is $(0, -5)$ and radius is 6 so $x^2 + (y+5)^2 = 36$

When $x = 0$, $y = -5 \pm \sqrt{36}$ so $y = 1$ or $y = -11$

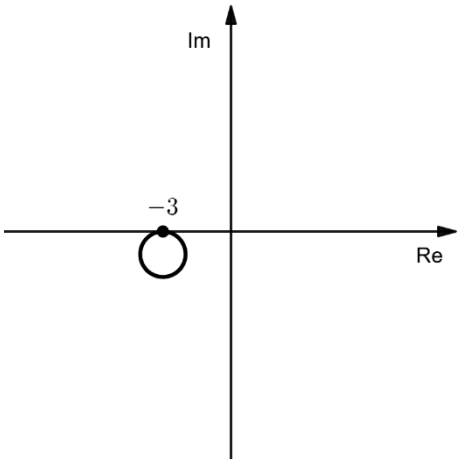
When $y = 0$, $x = \pm\sqrt{11}$



3) c)

Centre is $(-3, -1)$ and radius is 1 so $(x+3)^2 + (y+1)^2 = 1$

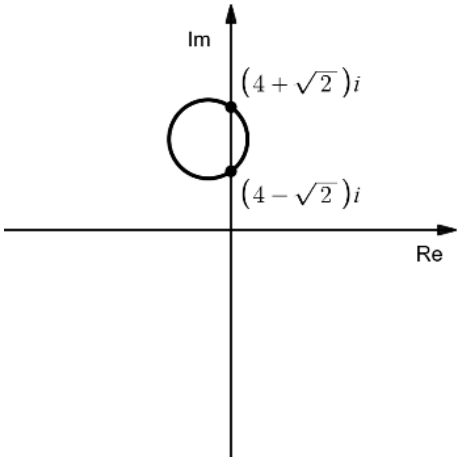
When $x = 0$, y has no real solutions, when $y = 0$, $x = -3$



3) d)

Centre is $(-1, 4)$ and radius is $\sqrt{3}$ so $(x+1)^2 + (y-4)^2 = 3$

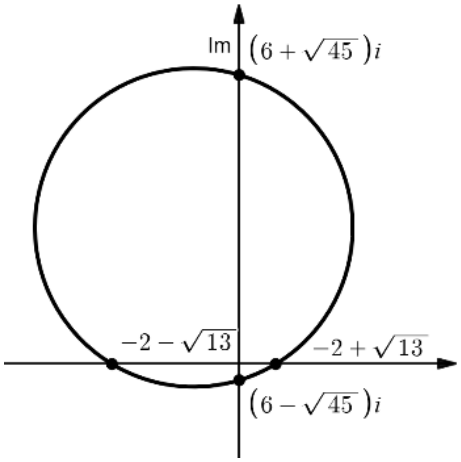
When $x = 0$, $y = 4 \pm \sqrt{2}$ and when $y = 0$, $x = \pm\sqrt{11}$



3) e)

Centre is $(-2, 6)$ and radius is 7 so $(x+2)^2 + (y-6)^2 = 49$

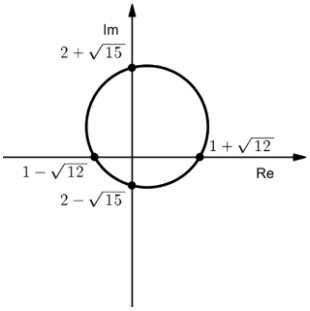
When $x = 0$, $y = 6 \pm \sqrt{45}$ and when $y = 0$, $x = -2 \pm \sqrt{13}$



3) f)

Centre is $(1, 2)$ and radius is 4 so $(x-1)^2 + (y-2)^2 = 16$

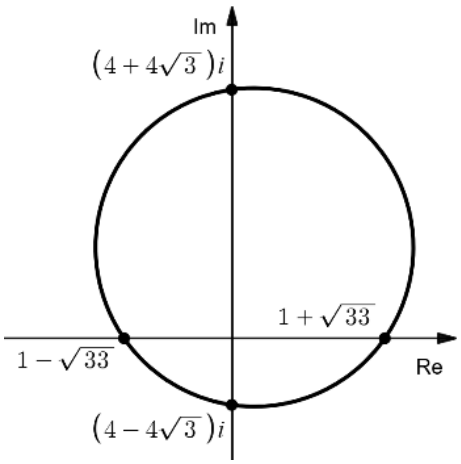
When $x = 0$, $y = 2 \pm \sqrt{15}$ and when $y = 0$, $x = 1 \pm \sqrt{12}$



4) a)

Centre is $(1, 4)$ and radius is 7 so $(x-1)^2 + (y-4)^2 = 49$

When $x = 0$, $y = 4 \pm 4\sqrt{3}$ and when $y = 0$, $x = 1 \pm \sqrt{33}$



4) b)

$|z - 1 - 4\mathrm{i}| = 7$ and $\mathrm{Re}(z) = 0$ when $z = \left(4 \pm 4\sqrt{3}\right)\mathrm{i}$

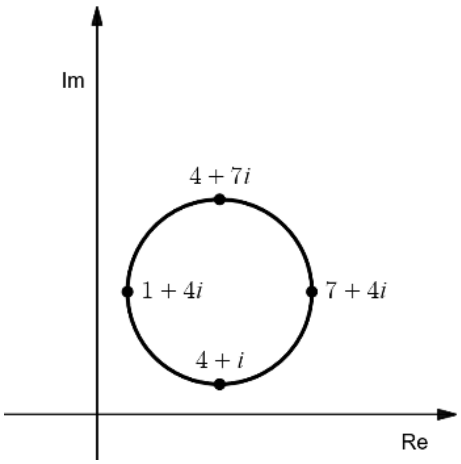
4) c)

$|z - 1 - 4\mathrm{i}| = 7$ and $\mathrm{Im}(z) = 0$ when $z = 1 \pm \sqrt{33}\mathrm{i}$

5) a)

Centre is $(4, 4)$ and radius is 3 so $(x - 4)^2 + (y - 4)^2 = 9$

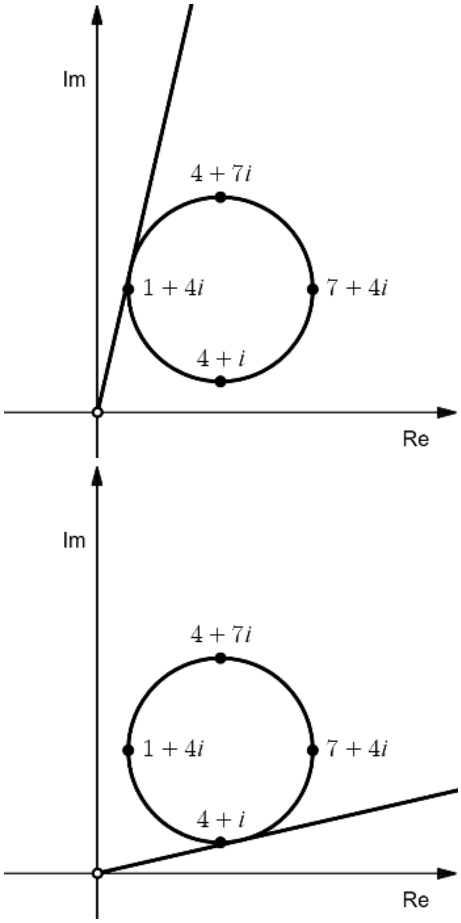
Since the radius is 3, the circle does not intersect with the axes, so fully lies within the top-right quadrant.



5) b)

$(x - 4)^2 + (y - 4)^2 = 9$

5) c)



Need to find when $y = mx$ and $(x-4)^2 + (y-4)^2 = 9$ have one point of intersection.

$(x-4)^2 + (mx-4)^2 = 9$

$x^2 - 8x + 16 + m^2x^2 - 8mx + 16 - 9 = 0$

$(m^2+1)x^2 - 8(1+m)x + 23 = 0$

A single point of intersection occurs when $b^2 - 4ac = 0$

$64(1+m)^2 - 92(m^2+1) = 0$

$64m^2 + 128m + 64 - 92m^2 - 92 = 0$

$-28m^2 + 128m - 28 = 0$

$\therefore m = \frac{16 \pm 3\sqrt{23}}{7}$

So the maximum $\arg z = \arctan\left(\frac{16+3\sqrt{23}}{7}\right) = 1.34$

and minimum $\arg z = \arctan\left(\frac{16-3\sqrt{23}}{7}\right) = 0.226$

5) d)

The maximum and minimum values of $|z|$ can be found by finding the distance from the origin to the centre, then adding or subtracting the radius:

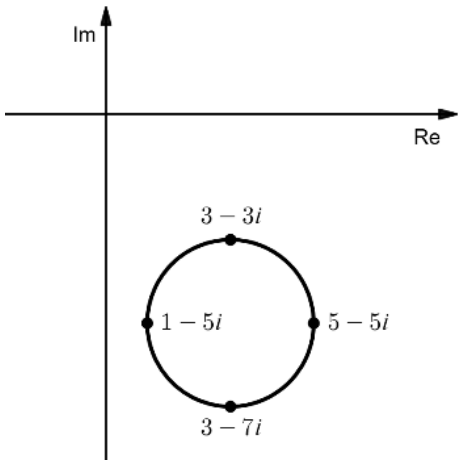
Maximum $|z| = 3 + \sqrt{4^2 + 4^2} = 3 + 4\sqrt{2}$

Minimum $|z| = 3 - \sqrt{4^2 + 4^2} = 3 - 4\sqrt{2}$

6) a)

Centre is $(3, -5)$ and radius is 2 so $(x - 3)^2 + (y + 5)^2 = 4$

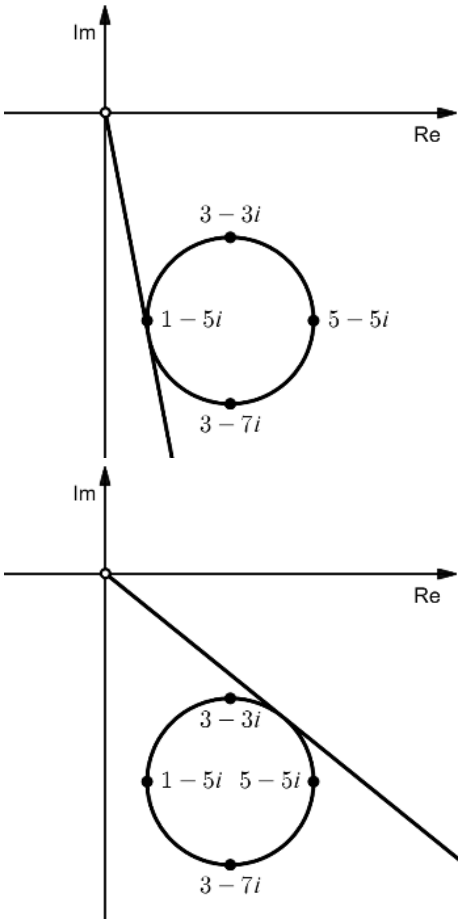
Since the radius is 2, the circle does not intersect with the axes, so fully lies within the top-right quadrant.



6) b)

$(x - 3)^2 + (y + 5)^2 = 4$

6) c)



Need to find when $y = mx$ and $(x-3)^2 + (y+5)^2 = 4$ have one point of intersection.

$(x-3)^2 + (mx+5)^2 = 4$

$x^2 - 6x + 9 + m^2x^2 + 10mx + 25 - 4 = 0$

$(m^2+1)x^2 + (10m-6)x + 30 = 0$

A single point of intersection occurs when $b^2 - 4ac = 0$

$(10m-6)^2 - 120(m^2+1) = 0$

$100m^2 - 120m + 36 - 120m^2 - 120 = 0$

$-20m^2 - 120m - 84 = 0$

$\therefore m = \frac{-15 \pm 2\sqrt{30}}{5}$

So the maximum $\arg z = \arctan\left(\frac{-15+2\sqrt{30}}{5}\right) = -0.680$ (3 SF)

and minimum $\arg z = \arctan\left(\frac{-15-2\sqrt{30}}{5}\right) = -1.38$ (3 SF)

6) d)

The maximum and minimum values of $|z|$ can be found by finding the distance from the origin to the centre, then adding or subtracting the radius:

Maximum $|z| = \sqrt{3^2 + (-5)^2} + 2 = 2 + \sqrt{34}$

Minimum $|z| = \sqrt{3^2 + (-5)^2} - 2 = -2 + \sqrt{34}$

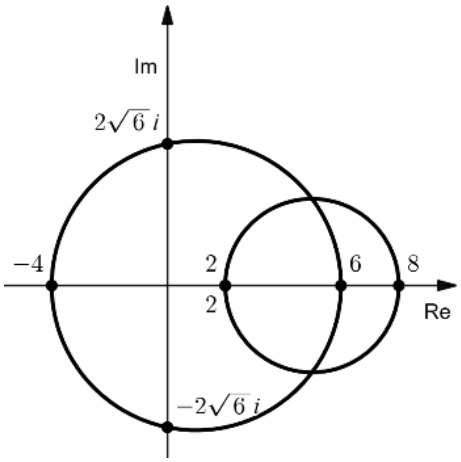
7) a)

Centre is $(1, 0)$ and radius is 5 so $(x-1)^2 + y^2 = 25$

When $x = 0$, $y = \pm 2\sqrt{6}$ and when $y = 0$, $x = -4, 6$

Centre is $(5, 0)$ and radius is 3 so $(x-5)^2 + y^2 = 9$

When $y = 0$, $x = 2, 8$



7) b)

Solve $(x-1)^2 + y^2 = 25$ and $(x-5)^2 + y^2 = 9$ simultaneously:

$(x-1)^2 + y^2 = 25$ gives $y^2 = 25 - (x-1)^2$

Subbing into the other equation:

$(x-5)^2 + 25 - (x-1)^2 = 9$

$x^2 - 10x + 25 + 25 - x^2 + 2x - 1 = 9$

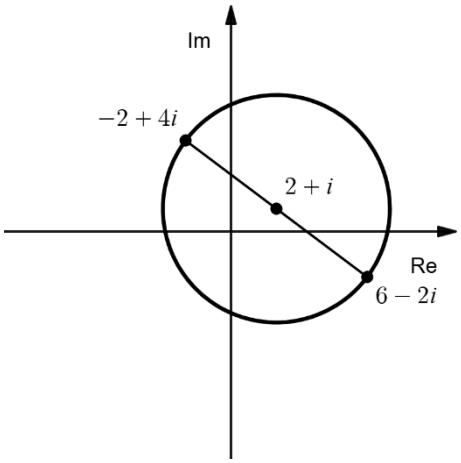
$8x + 40 = 0 \Rightarrow x = 5$

So $y = \pm 3$

Therefore, the two values of z are $5 + 3\text{i}$ and $5 - 3\text{i}$

8) a)

Midpoint is $2 + \mathrm{i}$ and radius is $\sqrt{4^2 + 3^2} = 5$



$|z - 2 - \mathrm{i}| = 5$

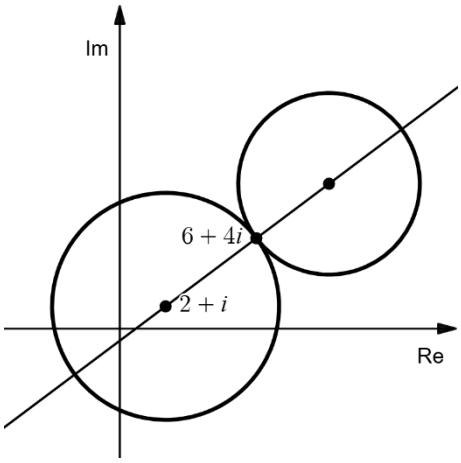
8) b)

$|z - p - q\mathrm{i}| = 4$ passes through the point $6 + 4\mathrm{i}$

$|6 - p + 4\mathrm{i} - q\mathrm{i}| = 2 \Rightarrow \sqrt{(6-p)^2 + (4-q)^2} = 4$

$\Rightarrow (6-p)^2 + (4-q)^2 = 16$

For the loci to meet the first loci once and not cross the axes, it must have a centre on the straight line passing through $2 + \mathrm{i}$ and $6 + 4\mathrm{i}$



Similar triangles: $\frac{4}{5} \times 4 = \frac{16}{5}$ and $\frac{3}{5} \times 4 = \frac{12}{5}$

$p = 2 + 4 + \frac{16}{5} = \frac{46}{5}$

$q = 1 + 3 + \frac{12}{5} = \frac{32}{5}$

Alternative Method:

Line passing through $2 + \mathrm{i}$ and $6 + 4\mathrm{i}$ in cartesian form: $y = \frac{3}{4}x - \frac{1}{2}$

So $q = \frac{3}{4}p - \frac{1}{2}$

$(6-p)^2 + \left(4 - \left(\frac{3}{4}p - \frac{1}{2}\right)\right)^2 = 16$

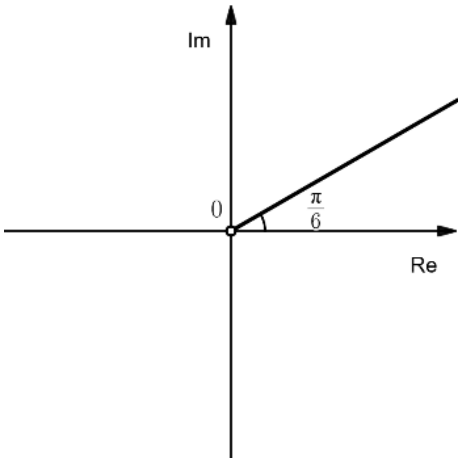
$(6-p)^2 + \left(\frac{9}{2} - \frac{3}{4}p\right)^2 = 16$

$36 - 12p + p^2 + \frac{81}{4} - \frac{27}{4}p + \frac{9}{16}p^2 = 16$

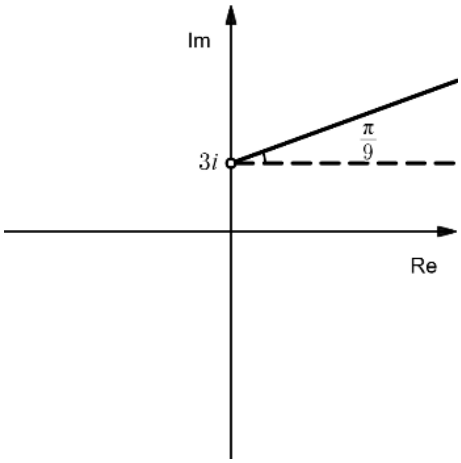
$\frac{25}{16}p^2 - \frac{75}{4}p + \frac{161}{4} = 0$ gives $p = \frac{14}{5}$ or $p = \frac{46}{5}$

so $p = \frac{46}{5}$ and $q = \frac{32}{5}$

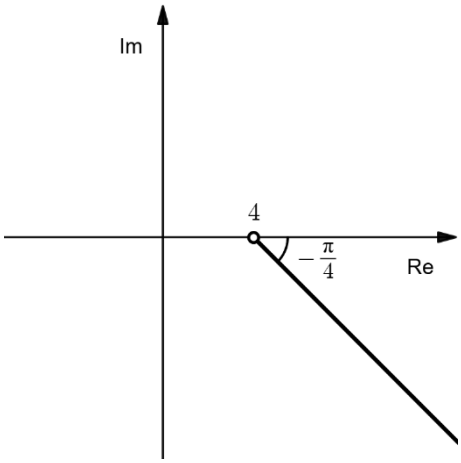
9) a)



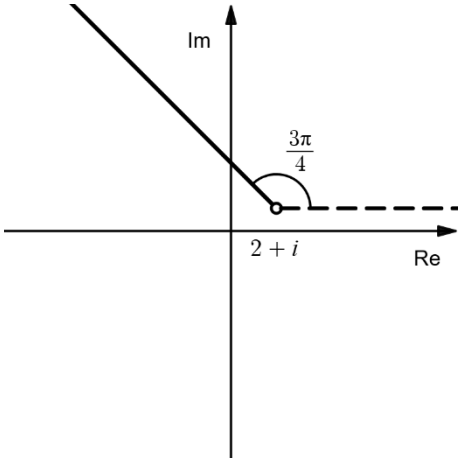
9) b)



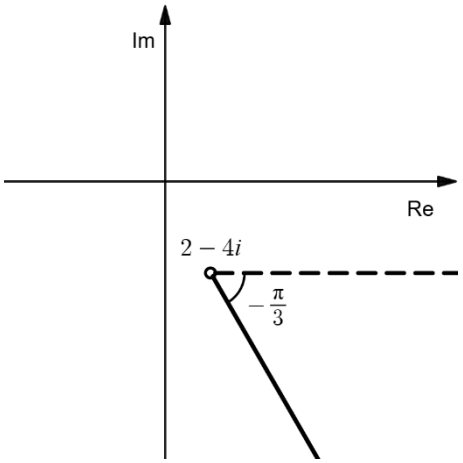
9) c)



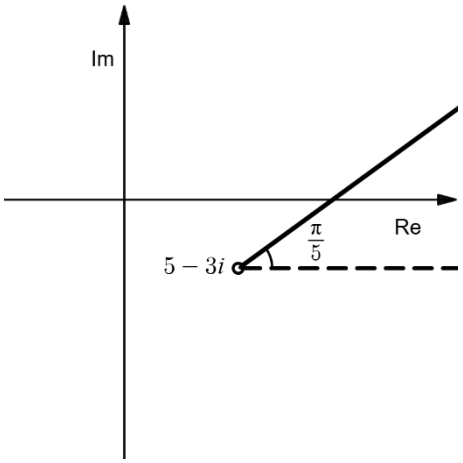
9) d)



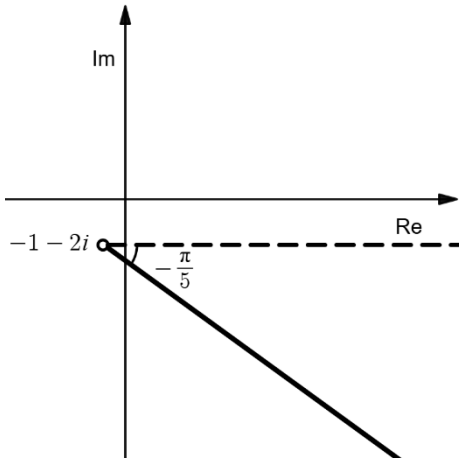
9) e)



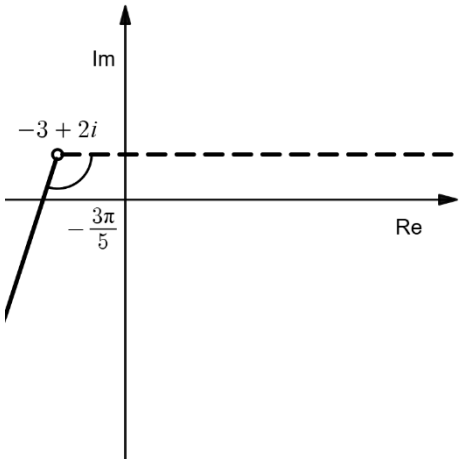
9) f)



9) g)



9) h)



10) a)

$\arctan(1) = \frac{\pi}{4}$

$\arg(z-2) = \frac{\pi}{4}$

10) b)

$\arctan(1) = \frac{\pi}{4}$ so that $\frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$

$\arg(z-3-\mathrm{i}) = \frac{3\pi}{4}$

10) c)

$\arctan(1) = \frac{\pi}{4}$

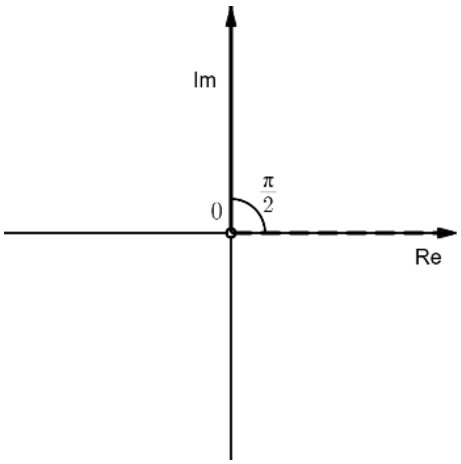
$\arg(z+2-2\mathrm{i}) = -\frac{\pi}{4}$

10) d)

$\arctan\left(\frac{4}{4\sqrt{3}}\right) = \frac{\pi}{6}$ so that $-\frac{\pi}{2} - \frac{\pi}{6}$

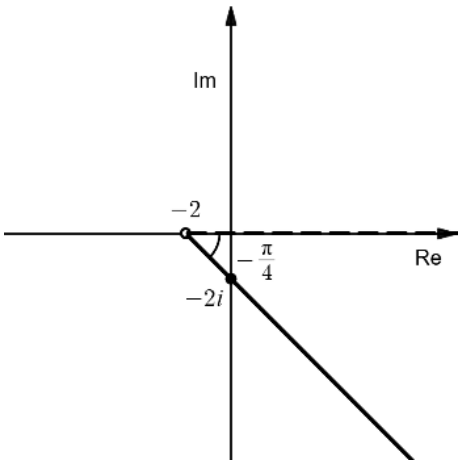
$\arg(z - 2 - \mathrm{i}) = -\frac{2\pi}{3}$

11) a)



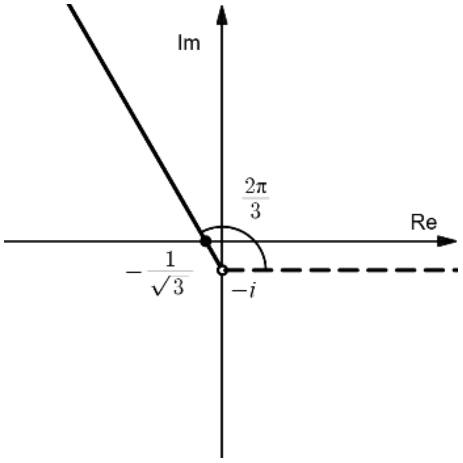
$x = 0$ for $y > 0$

11) b)



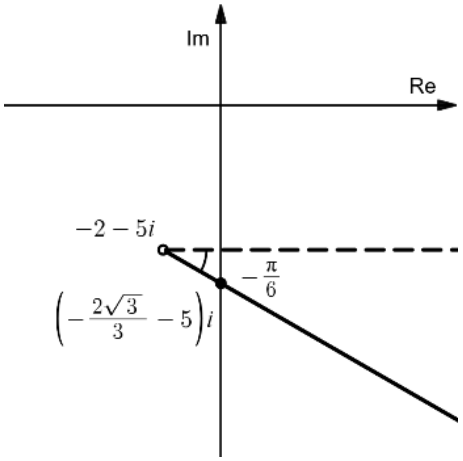
$y + 2 = \tan\left(-\frac{\pi}{4}\right)(x - 0)$ so $y = -x - 2$ for $x > -2$

11) c)



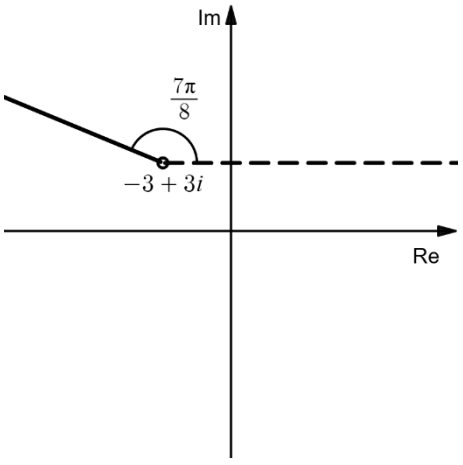
$y + 1 = \tan\left(\frac{2\pi}{3}\right)(x - 0)$ so $y = -\sqrt{3}x - 1$ for $x < 0$

11) d)



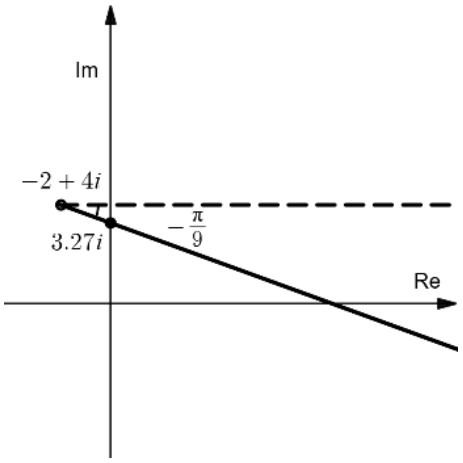
$y + 5 = \tan\left(-\frac{\pi}{6}\right)(x + 2)$ so $y = -\frac{\sqrt{3}}{3}x - \frac{2\sqrt{3}}{3} - 5 +$ for $x > -2$

11) e)



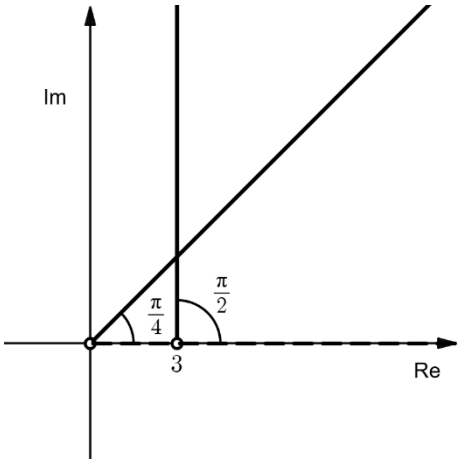
$y - 3 = \tan\left(\frac{7\pi}{8}\right)(x + 3)$ so $y = -0.414x + 1.76$ for $x < -3$ (3 SF)

11) f)



$y-4=\tan\left(-\frac{\pi}{9}\right)(x+2)$ so $y=-0.364x+3.27$ for $x>-2$ (3 SF)

12) a)



12) b)

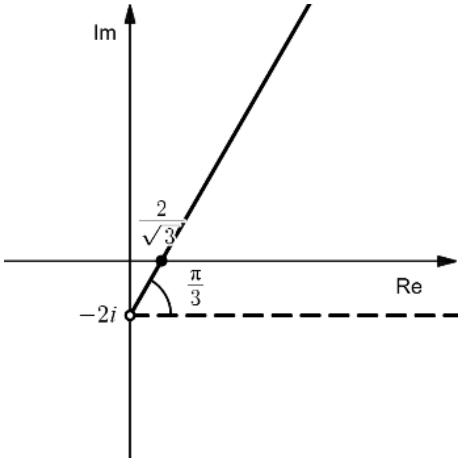
$y=x$ and $x=3$ intersect at $(3,3)$ so $z=3+3\text{i}$

12) c)

$\arg(3+3\text{i}-2\text{i})=\arg(3+\text{i})=\arctan\left(\frac{1}{3}\right)=0.322$ (3 SF)

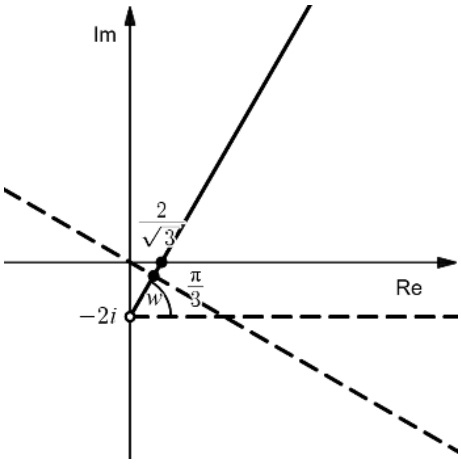
13) a)

$y + 2 = \tan\left(\frac{\pi}{3}\right)(x - 0) \Rightarrow y = \sqrt{3}x - 2$



13) b)

$y = \sqrt{3}x - 2$ gives a perpendicular line of $y = -\frac{1}{\sqrt{3}}x$



w is the point that $y = \sqrt{3}x - 2$ and $y = -\frac{1}{\sqrt{3}}x$ intersect:

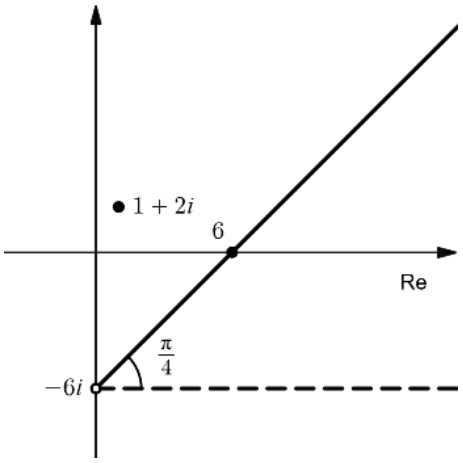
$\sqrt{3}x - 2 = -\frac{1}{\sqrt{3}}x \Rightarrow x = \frac{\sqrt{3}}{2}$ and $y = -\frac{1}{2}$

Therefore $w = \frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}$

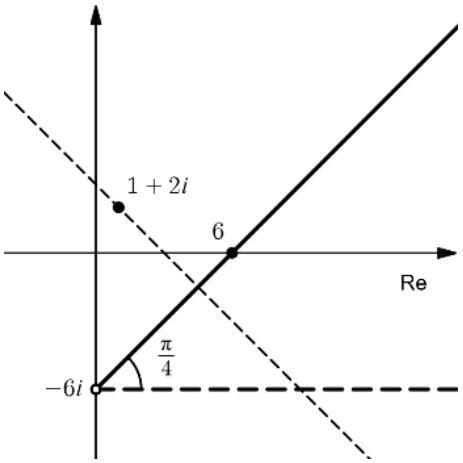
13) c)

$|z| = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} = 1$

14) a)



14) b)



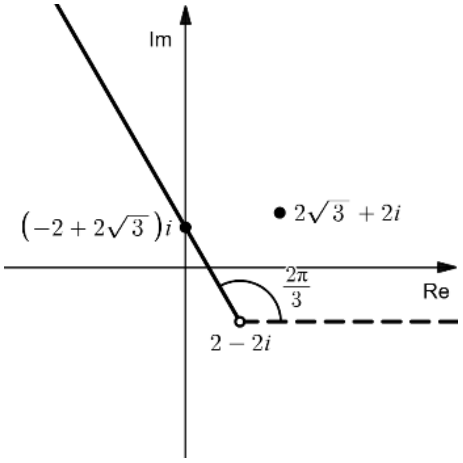
Since $y + 6 = \tan\left(\frac{\pi}{4}\right)(x - 0) \Rightarrow y = x - 6$ is the cartesian line of the argument, the perpendicular line which passes through $(1,2)$ is given by

$y - 2 = -(x - 1) \Rightarrow y = -x + 3$

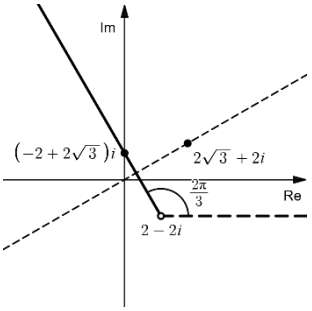
These lines intersect at $\left(\frac{9}{2}, -\frac{3}{2}\right)$

Shortest distance between w and $\left(\frac{9}{2}, -\frac{3}{2}\right)$ is $\sqrt{\left(\frac{9}{2} - 1\right)^2 + \left(-\frac{3}{2} - 2\right)^2} = \frac{7}{2}\sqrt{2}$

15) a)



15) b)



Since $y+2=\tan\left(\frac{2\pi}{3}\right)(x-2) \Rightarrow y=-\sqrt{3}x+2\sqrt{3}-2$ is the cartesian line of the argument, the perpendicular line which passes through $\left(2\sqrt{3},2\right)$ is given by

$y-2=\frac{1}{\sqrt{3}}(x-2\sqrt{3}) \Rightarrow y=\frac{1}{\sqrt{3}}x$

These lines intersect at $\left(\frac{3}{2}-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}-\frac{1}{2}\right)$

Shortest distance between w and $\left(\frac{3}{2}-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}-\frac{1}{2}\right)$ is

$$\sqrt{\left(2\sqrt{3}-\frac{3}{2}+\frac{\sqrt{3}}{2}\right)^2+\left(2-\frac{\sqrt{3}}{2}+\frac{1}{2}\right)^2}=\sqrt{\left(\frac{5}{2}\sqrt{3}-\frac{3}{2}\right)^2+\left(\frac{5}{2}-\frac{\sqrt{3}}{2}\right)^2}$$

$$=\sqrt{21-\frac{15}{2}\sqrt{3}+7-\frac{5}{2}\sqrt{3}}=\sqrt{28-10\sqrt{3}}$$

$\sqrt{28-10\sqrt{3}}=a+b\sqrt{3} \Rightarrow 28-10\sqrt{3}=\left(a+b\sqrt{3}\right)^2=a^2+2ab\sqrt{3}+3b^2$

$2ab=-10$ and $a^2+3b^2=28$

Solving simultaneously gives

$b=-\frac{5}{a} \Rightarrow a^2+3\left(-\frac{5}{a}\right)^2=28 \Rightarrow a^4+75=28a^2 \Rightarrow a^4-28a^2+75=0$

$\therefore a^2=25$ or $a^2=3$ but a is an integer so $a=\pm5$

So $a=\pm5$, $b=\mp1$

Since it is a distance $a=5, b=-1$ so $5-\sqrt{3}$

16) a)

Points: 4 and i gives $|z-4|=|z-\mathrm{i}|$

Midpoint: $\left(2,\frac{1}{2}\right)$ and gradient between points is $-\frac{1}{4}$

$\therefore y-\frac{1}{2}=4(x-2)\Rightarrow y=4x-\frac{15}{2}$

16) b)

Points: 2 and $-3\mathrm{i}$ gives $|z-2|=|z+3\mathrm{i}|$

Midpoint: $\left(1,-\frac{3}{2}\right)$ and gradient between points is $\frac{3}{2}$

$\therefore y+\frac{3}{2}=-\frac{2}{3}(x-1)\Rightarrow y=-\frac{2}{3}x-\frac{5}{6}$

16) c)

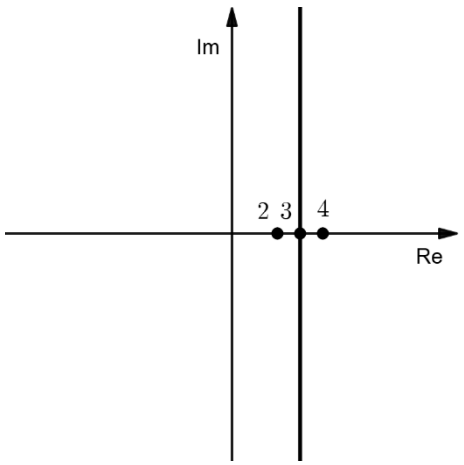
Points: $4-2\mathrm{i}$ and $2-4\mathrm{i}$ gives $|z-4+2\mathrm{i}|=|z-2+4\mathrm{i}|$

Midpoint: $(3,-3)$ and gradient between points is 1

$\therefore y+3=-1(x-3)\Rightarrow y=-x$

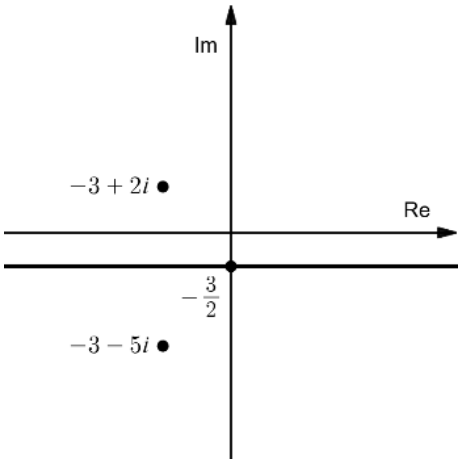
17) a)

$x=3$



17) b)

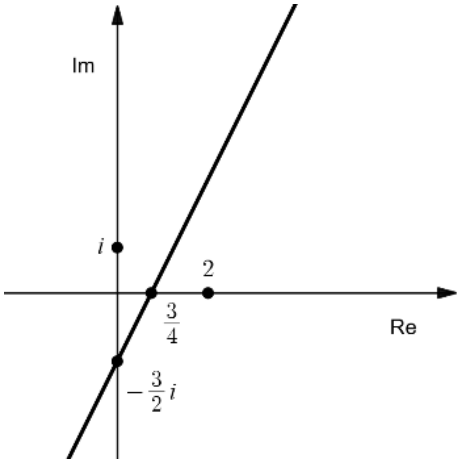
$y = -\frac{3}{2}$



17) c)

Points: 2 and i gives midpoint: $\left(1, \frac{1}{2}\right)$ and gradient between points is $-\frac{1}{2}$

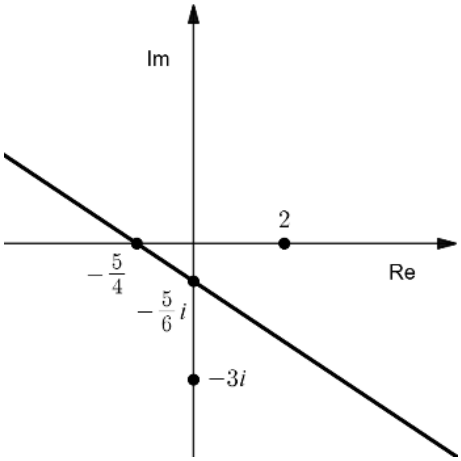
$\therefore y - \frac{1}{2} = 2(x-1) \Rightarrow y = 2x - \frac{3}{2}$



17) d)

Points: 2 and $-3\mathrm{i}$ gives midpoint: $\left(1, -\frac{3}{2}\right)$ and gradient between points is $\frac{3}{2}$

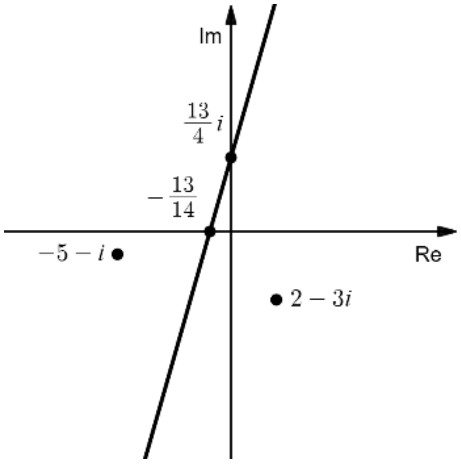
$\therefore y + \frac{3}{2} = -\frac{2}{3}(x-1) \Rightarrow y = -\frac{2}{3}x - \frac{5}{6}$



17) e)

Points: $-5-\mathrm{i}$ and $2-3\mathrm{i}$ gives midpoint: $\left(-\frac{3}{2}, -2\right)$ and gradient between points is $-\frac{2}{7}$

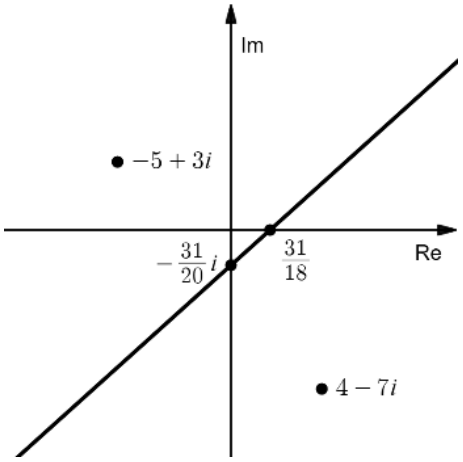
$\therefore y + 2 = \frac{7}{2}\left(x + \frac{3}{2}\right) \Rightarrow y = \frac{7}{2}x + \frac{13}{4}$



17) f)

Points: $-5+3\text{i}$ and $4-7\text{i}$ gives midpoint: $\left(-\frac{1}{2},-2\right)$ and gradient between points is $-\frac{10}{9}$

$\therefore y+2=\frac{9}{10}\left(x+\frac{1}{2}\right)\Rightarrow y=\frac{9}{10}x-\frac{31}{20}$



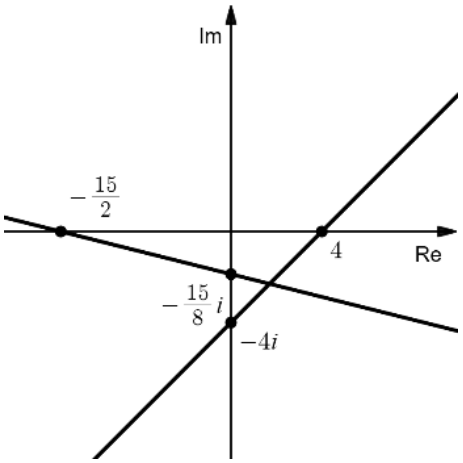
18) a)

Points: $2-\text{i}$ and $3-2\text{i}$ gives midpoint: $\left(\frac{5}{2},-\frac{3}{2}\right)$ and gradient between points is -1

$\therefore y+\frac{3}{2}=1\left(x-\frac{5}{2}\right)\Rightarrow y=x-4$

Points: 1 and -4i gives midpoint: $\left(\frac{1}{2},-2\right)$ and gradient between points is 4

$\therefore y+2=-\frac{1}{4}\left(x-\frac{1}{2}\right)\Rightarrow y=-\frac{1}{4}x-\frac{15}{8}$



18) b)

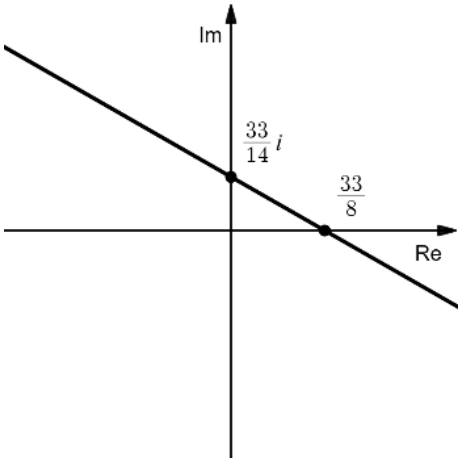
$y=x-4$ and $y=-\frac{1}{4}x-\frac{15}{8}$ gives $x-4=-\frac{1}{4}x-\frac{15}{8}$ gives $x=\frac{17}{10}$ and $y=-\frac{23}{10}$

so $\frac{17}{10}-\frac{23}{10}\text{i}$

19) a)

Points: -4 and $7i$ gives midpoint: $\left(-2,\frac{7}{2}\right)$ and gradient between points is $\frac{7}{4}$

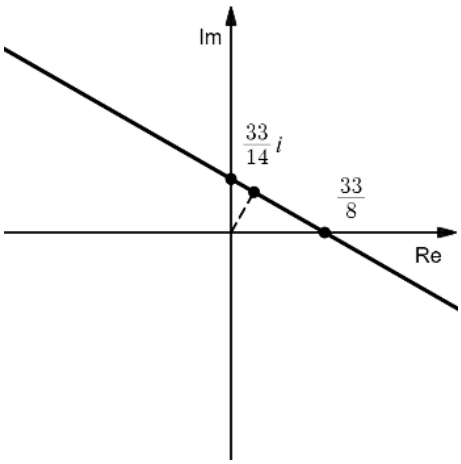
$\therefore y-\frac{7}{2}=-\frac{4}{7}(x+2) \Rightarrow y=-\frac{4}{7}x+\frac{33}{14}$



19) b)

$y=-\frac{4}{7}x+\frac{33}{14}$

19) c)



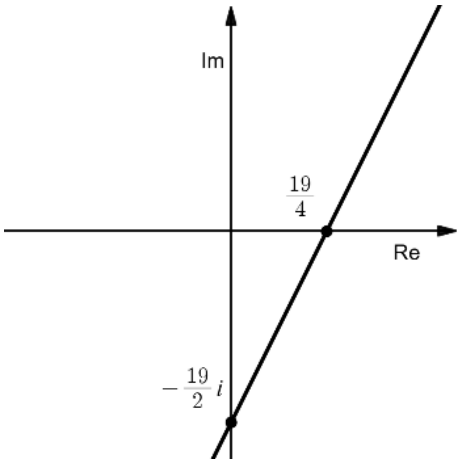
$y=-\frac{4}{7}x+\frac{33}{14}$ and $y=\frac{7}{4}x$ intersect when $-\frac{4}{7}x+\frac{33}{14}=\frac{7}{4}x$ which gives $\left(\frac{66}{65},\frac{231}{130}\right)$

$\sqrt{\left(\frac{66}{65}\right)^2+\left(\frac{231}{130}\right)^2}=\frac{33}{130}\sqrt{65}$

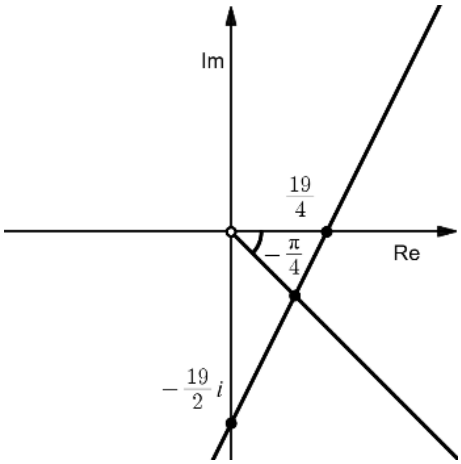
20) a)

Points: $2-3\mathrm{i}$ and $4-4\mathrm{i}$ gives midpoint: $\left(3,-\frac{7}{2}\right)$ and gradient between points is $-\frac{1}{2}$

$\therefore y+\frac{7}{2}=2(x-3) \Rightarrow y=2x-\frac{19}{2}$



20) b)



$\arg(z)=\frac{\pi}{4}$ has cartesian equation $y=-x$

$-x=2x-\frac{19}{2}$ gives $x=\frac{19}{6}$

So z is $\frac{19}{6}-\frac{19}{6}\mathrm{i}$

21) a)

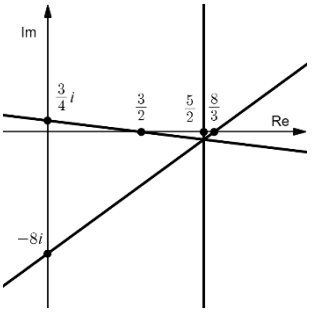
Points: -1 and 6 gives midpoint: $\left(0, \frac{5}{2}\right)$ so $x = \frac{5}{2}$

Points: $2\mathrm{i}$ and 6 gives midpoint: $(3, 1)$ and gradient between points is $-\frac{1}{3}$

$\therefore y - 1 = 3(x - 3) \Rightarrow y = 3x - 8$

Points: -1 and $2\mathrm{i}$ gives midpoint: $\left(-\frac{1}{2}, 1\right)$ and gradient between points is 2

$\therefore y - 1 = -\frac{1}{2}\left(x + \frac{1}{2}\right) \Rightarrow y = -\frac{1}{2}x + \frac{3}{4}$



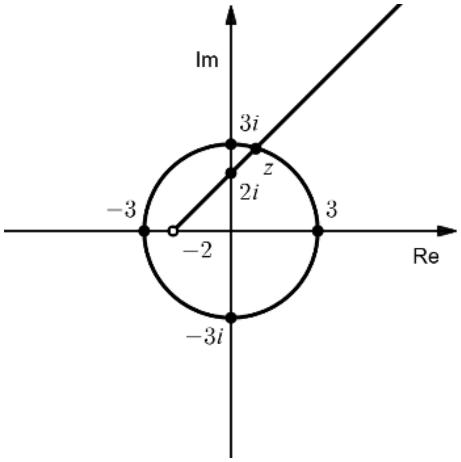
21) b)

$x = \frac{5}{2}$ and $y = 3x - 8$ and $y = -\frac{1}{2}x + \frac{3}{4}$

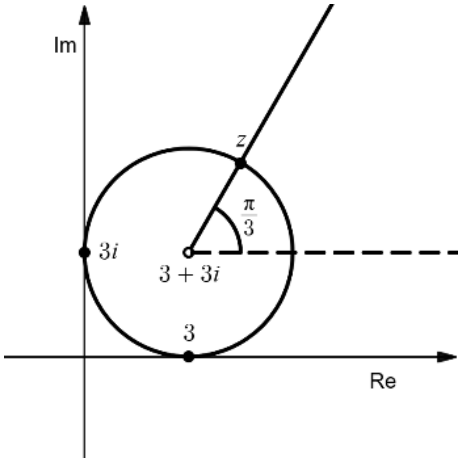
$x = \frac{5}{2}$ gives $y = 3 \times \frac{5}{2} - 8 = -\frac{1}{2}$

So $z = \frac{5}{2} - \frac{1}{2}\mathrm{i}$

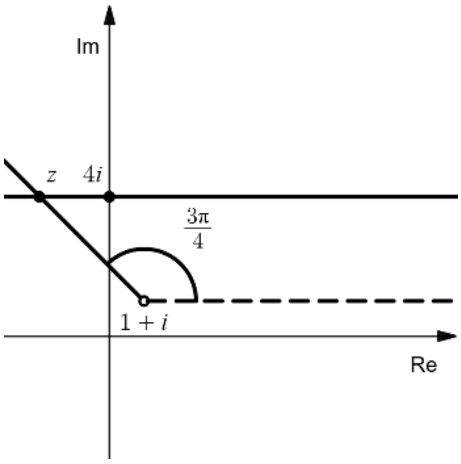
22) a)



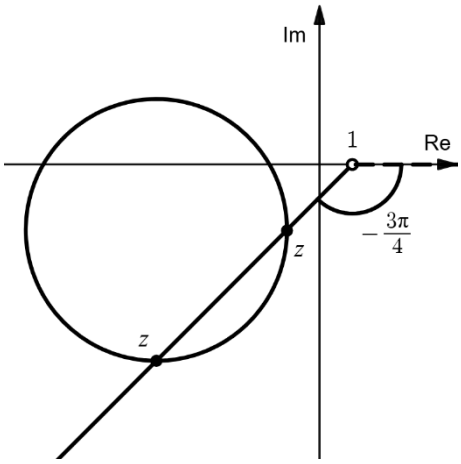
22) b)


Im
z
$\frac{\pi}{3}$
$3i$
$3+3i$
3
Re

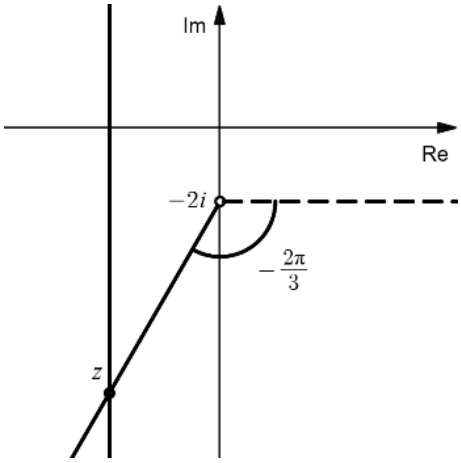
22) c)


Im
z
$4i$
$\frac{3\pi}{4}$
$1+i$
Re

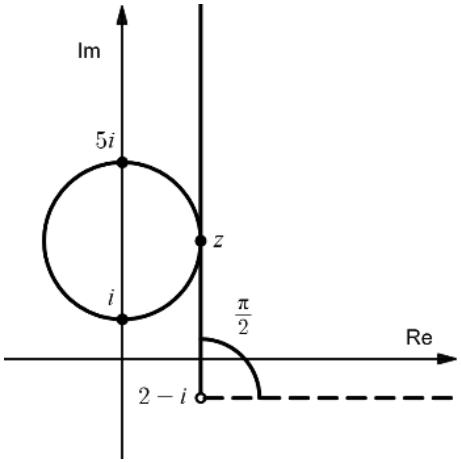
22) d)


Im
1
Re
$-\frac{3\pi}{4}$
z
z

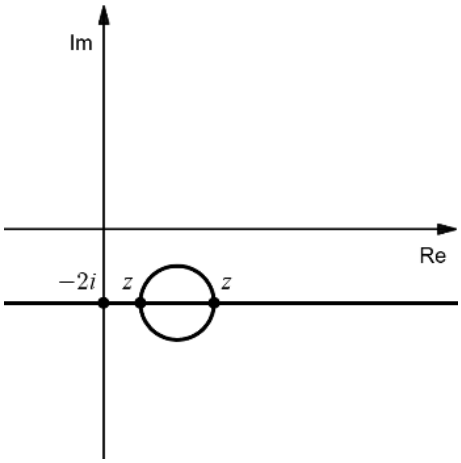
22) e)



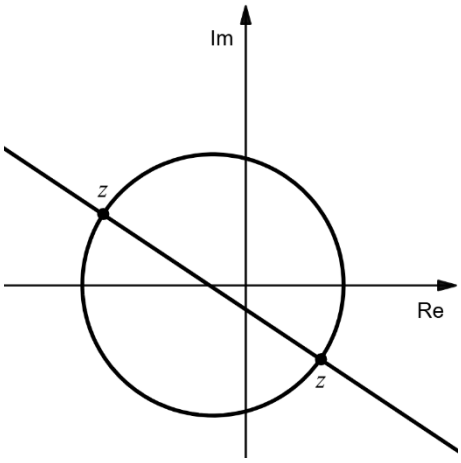
22) f)



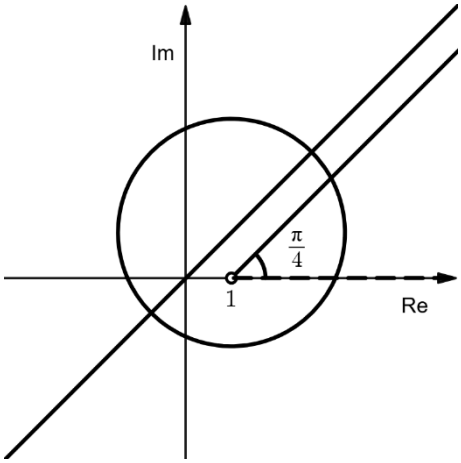
22) g)



22) h)

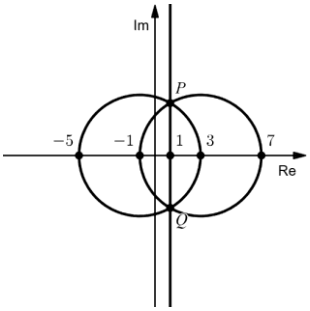


22) i)



There are no values of z for which all the conditions hold.

23)



24) a)

Centre at 1, radius of 5 so $|z-1|=5$

24) b)

Angle is $\arctan(1) = \frac{\pi}{4}$ so $\arg(z+2) = \frac{\pi}{4}$

24) c)

Solve $(x-1)^2 + y^2 = 5^2$ and $y = x + 2$ simultaneously:

$(x-1)^2 + (x+2)^2 = 5^2$

$x^2 - 2x + 1 + x^2 + 4x + 4 = 25$

$2x^2 + 2x - 20 = 0$ which gives $x = -\frac{1}{2} \pm \frac{\sqrt{41}}{2}$

Choosing the positive solution:

$x = -\frac{1}{2} + \frac{\sqrt{41}}{2}$ and $y = \frac{3}{2} \pm \frac{\sqrt{41}}{2}$

$\therefore z = -\frac{1}{2} + \frac{\sqrt{41}}{2} + \left(\frac{3}{2} + \frac{\sqrt{41}}{2}\right)\mathrm{i}$

25) a)

Centre at $1 - 3\mathrm{i}$, radius of 3 so $|z - 1 + 3\mathrm{i}| = 3$

25) b)

$|z + 4 - 3\mathrm{i}| = |z + 2 - 4\mathrm{i}|$

25) c)

$y - \frac{7}{2} = -2(x+3) \Rightarrow y = -2x - \frac{5}{2}$ and $(x-1)^2 + (y+3)^2 = 9$

Solve $y = -2x - \frac{5}{2}$ and $(x-1)^2 + (y+3)^2 = 9$ simultaneously:

$(x-1)^2 + \left(-2x - \frac{5}{2} + 3\right)^2 = 9$

$(x-1)^2 + \left(-2x + \frac{1}{2}\right)^2 = 9$

$x^2 - 2x + 1 + 4x^2 - 2x + \frac{1}{4} = 9$

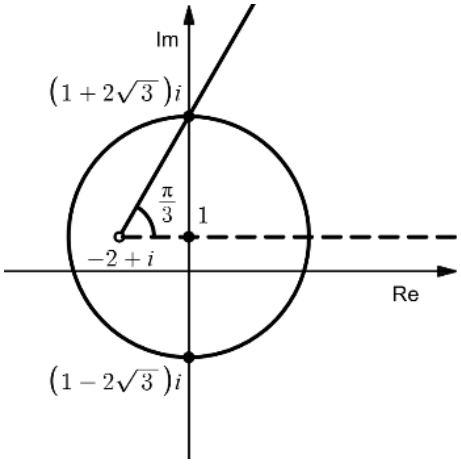
$5x^2 - 4x - \frac{31}{4} = 0$ which gives $x = \frac{2}{5} \pm \frac{3}{10}\sqrt{19}$ and $y = -\frac{33}{10} \mp \frac{3}{5}\sqrt{19}$

$\frac{2}{5} + \frac{3}{10}\sqrt{19} + \left(-\frac{33}{10} - \frac{3}{5}\sqrt{19}\right)\mathrm{i}$ and $\frac{2}{5} - \frac{3}{10}\sqrt{19} + \left(-\frac{33}{10} + \frac{3}{5}\sqrt{19}\right)\mathrm{i}$

26)

Circles crosses imaginary axis at $\left(1+2\sqrt{3}\right)$i and $\left(1-2\sqrt{3}\right)$i

$2\tan\left(\frac{\pi}{3}\right)=2\sqrt{3}$ so the line passes through at $1+2\sqrt{3}$



$z=\left(1+2\sqrt{3}\right)$i

27)

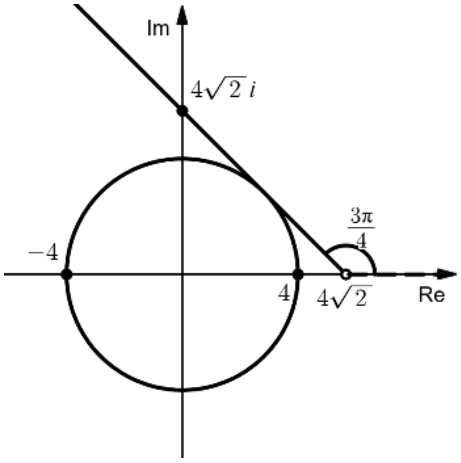
Circle is centred at origin, with radius $4 \Rightarrow x^2+y^2=16$

Cartesian equation of the half line: $y-0=-\left(x-4\sqrt{2}\right) \Rightarrow y=-x+4\sqrt{2}$

$x^2+\left(-x+4\sqrt{2}\right)^2=16$

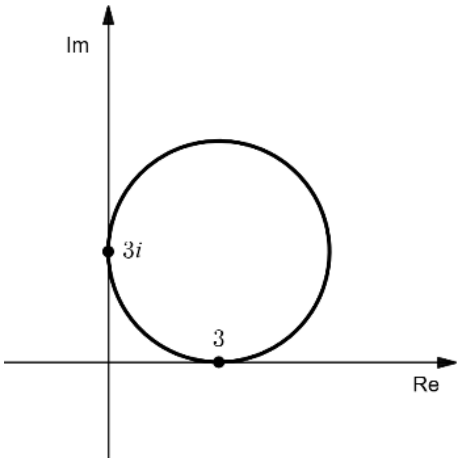
$x^2+x^2-8\sqrt{2}x+32=16$

$2x^2-8\sqrt{2}x+16=0$ which gives $x=2\sqrt{2}$ and $y=2\sqrt{2}$



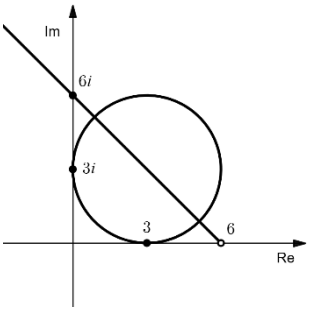
$z=2\sqrt{2}+2\sqrt{2}$i

28) a)



28) b)

$\arg(z-6)=\frac{3\pi}{4}$ is given by $y=-x+6$ for $x<3$



Solve $y=-x+6$ and $(x-3)^2+(y-3)^2=9$ simultaneously:

$(x-3)^2+(-x+6\ -3)^2=9$

$(x-3)^2+(-x+3)^2=9$

$x^2-6x+9+x^2-6x+9=9$

$2x^2-12x+9=0 \Rightarrow x=3\pm\frac{3\sqrt{2}}{2}$ and $y=3\mp\frac{3\sqrt{2}}{2}$

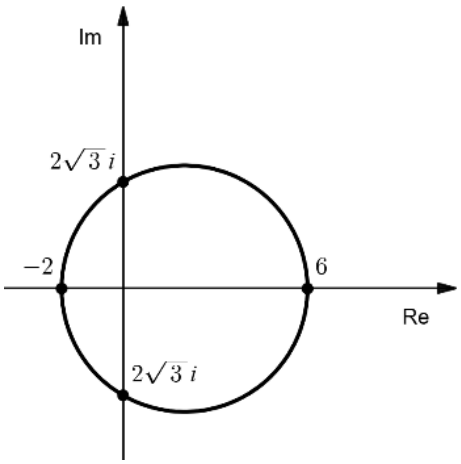
$\sqrt{\left(3+\frac{3\sqrt{2}}{2}\right)^2+\left(3-\frac{3\sqrt{2}}{2}\right)^2}=3\sqrt{3}$

$|z|=3\sqrt{3}$

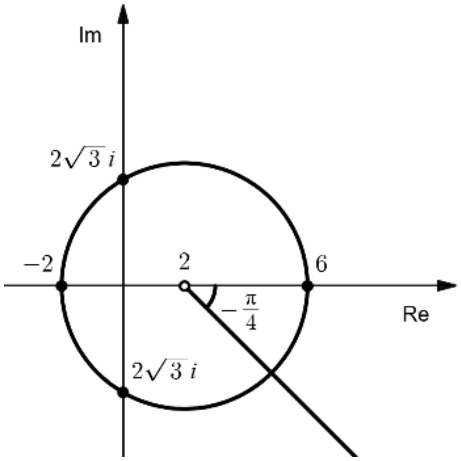
29) a)

$(x-2)^2+y^2=16$

When $x=0, y=\sqrt{12}=2\sqrt{2}$



29) b)



Solve $(x-2)^2+y^2=16$ and $y=-x+2$ for $x>2$ simultaneously:

$(x-2)^2+(-x+2)^2=16$

$x^2-4x+4+x^2-4x+4=16$

$2x^2-8x-8=0$ gives $x=2\pm 2\sqrt{2}$

So $x=2+2\sqrt{2}$ and $y=-2\sqrt{2}$

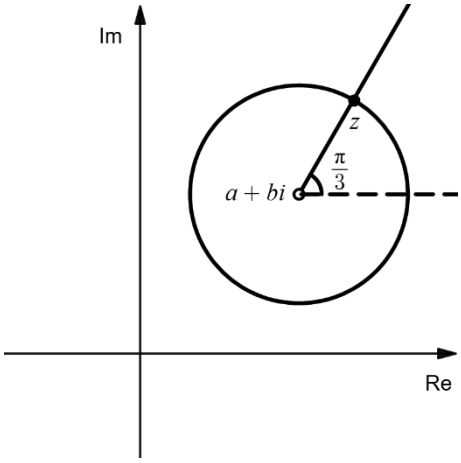
$\therefore z=2+2\sqrt{2}-2\sqrt{2}\mathrm{i}$

29) c)

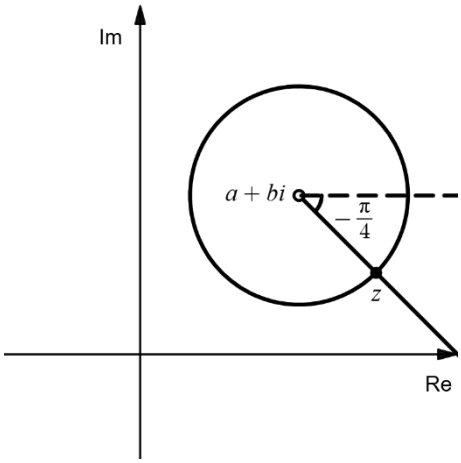
$\left(2+2\sqrt{2}\right)^2+\left(-2\sqrt{2}\right)^2=20+8\sqrt{2}$

$|z|^2=20+8\sqrt{2}$

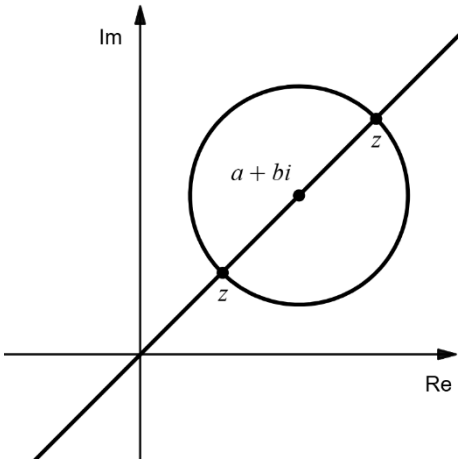
30) a)



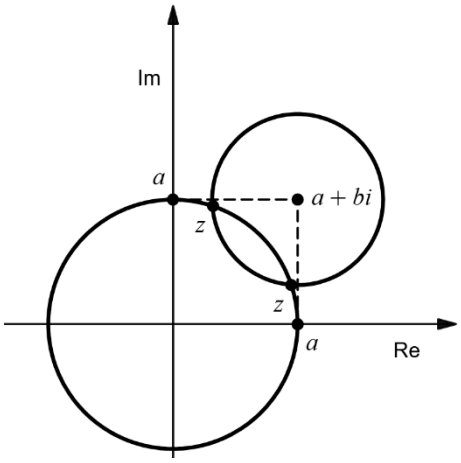
30) b)



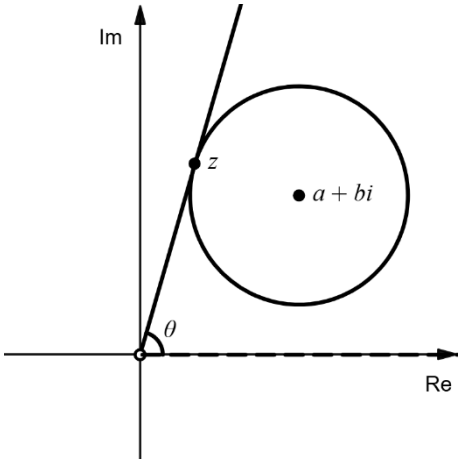
30) c)



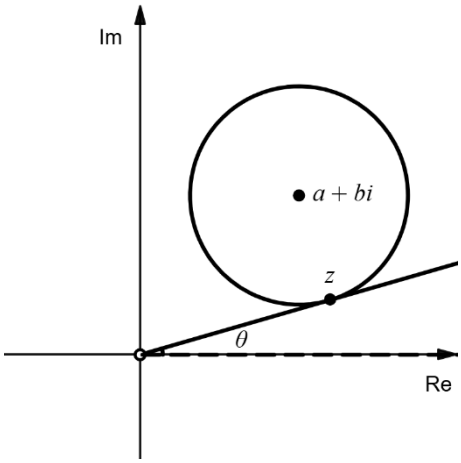
30) d)



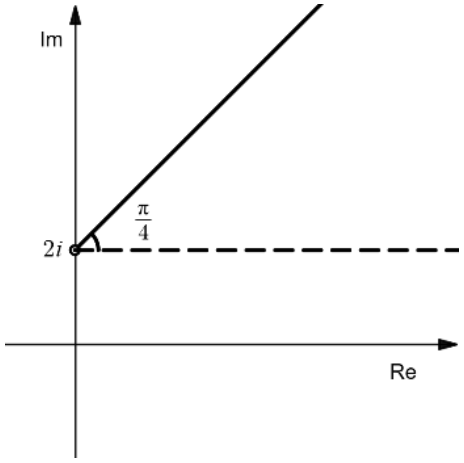
30) e)



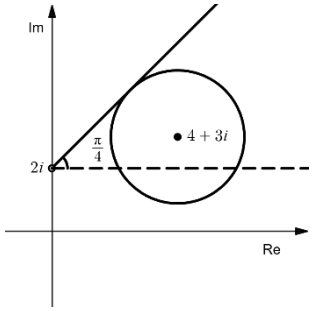
30) f)



31) a)



31) b)



Need the radius for which the line touches the circle.

$(x-4)^2+(y-3)^2=k^2$ and $y=x+2$

$(x-4)^2+(x+2-3)^2=k^2$

$(x-4)^2+(x-1)^2=k^2$

$x^2-8x+16+x^2-2x+1-k^2=0$

$2x^2-10x+17-k^2=0$

Consider $b^2-4ac=0$:

$(-10)^2-4\times2\times(17-k^2)=0$

$8k^2-36=0$

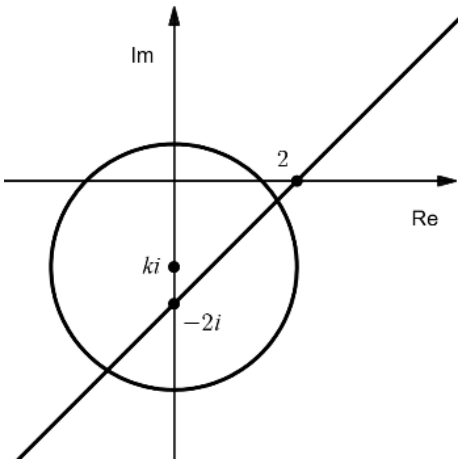
$k^2=\frac{9}{2}\Rightarrow k=\pm\frac{3}{\sqrt{2}}=\pm\frac{3\sqrt{2}}{2}$

Since k is positive, $k=\frac{3\sqrt{2}}{2}$

$\therefore |z-4-3\mathrm{i}|=\frac{3\sqrt{2}}{2}$

32)

As per the diagram:



There must be solution(s) to $x^2 + (y-k)^2 = 4$ and $y = x - 2$

$x^2 + (x-2-k)^2 = 4$

$x^2 + x^2 - 4x - 2kx + k^2 + 4k + 4 = 4$

$2x^2 - (4+2k)x + (4k+k^2) = 0$

Consider $b^2 - 4ac = 0$:

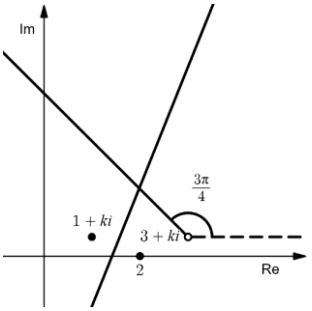
$(4+2k)^2 - 4 \times 2(4k+k^2) = 0$

$16 + 16k + 4k^2 - 32k - 8k^2 = 0$

$k^2 + 4k - 4 = 0$ giving $k = -2 \pm 2\sqrt{2}$

$\therefore -2 - 2\sqrt{2} \leq k \leq -2 + 2\sqrt{2}$

33)



Midpoint: $\left(\frac{3}{2},\frac{k}{2}\right)$ and gradient: $-\frac{k}{1}$

$y-\frac{k}{2}=\frac{1}{k}\left(x-\frac{3}{2}\right)\Rightarrow y=\frac{1}{k}x-\frac{3}{2k}+\frac{k}{2}$

Since k is positive, the gradient of the line is positive.

We need the point where the line passes through the point $3+k\mathrm{i}$

$k=\frac{1}{k}\times 3-\frac{3}{2k}+\frac{k}{2}$

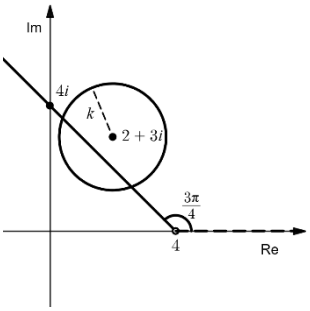
$k=\frac{6}{2k}-\frac{3}{2k}+\frac{k^2}{2k}$

$2k^2=3+k^2$

$k^2=3\therefore k=\sqrt{3}$

So that the interval is $0<k<\sqrt{3}$

34)



Need the values of k for which the line intersects with the circle.

$(x-2)^2+(y-3)^2=k^2$ and $y=-x+4$

$(x-2)^2+(-x+4-3)^2=k^2$

$(x-2)^2+(-x+1)^2=k^2$

$x^2-4x+4+x^2-2x+1=k^2$

$2x^2-6x+5-k^2=0$

Consider $b^2-4ac=0$:

$(-6)^2-4\times2\times(5-k^2)=0$

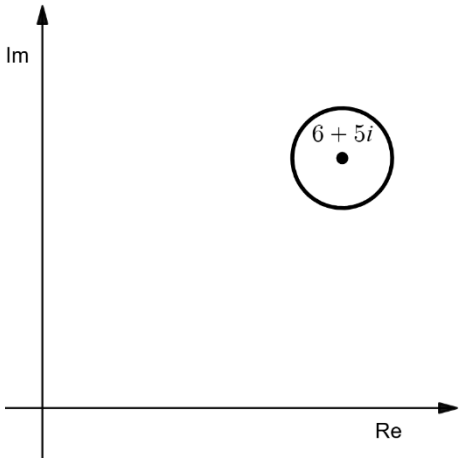
$36-40+8k^2=0$

$k^2=\frac{1}{2}\Rightarrow k=\pm\sqrt{\frac{1}{2}}=\pm\frac{\sqrt{2}}{2}$

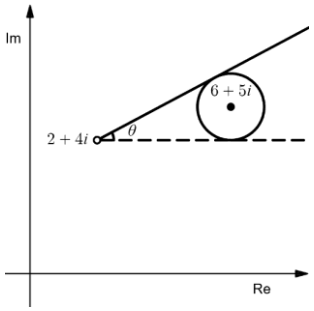
Since k is positive, $k=\frac{\sqrt{2}}{2}$

$\therefore k\geq\frac{\sqrt{2}}{2}$

35) a)



35) b)



Need the values of θ for which the line intersects with the circle exactly once.

$(x-6)^2+(y-5)^2=1$ and $y-4=m(x-2)$ where $m=\tan\theta$

$(x-6)^2+(mx-2m+4-5)^2=1$

$(x-6)^2+(mx-2m-1)^2=1$

$x^2-12x+36+m^2x^2-4m^2x+4m^2-2mx+4m+1=1$

$(1+m^2)x^2+(-4m^2-2m-12)x+(4m^2+4m+36)=0$

Consider $b^2-4ac=0$:

$(-4m^2-2m-12)^2-4(1+m^2)(4m^2+4m+36)=0$

$16m^4+16m^3+100m^2+48m+144-16m^4-16m^3-160m^2-16m-144=0$

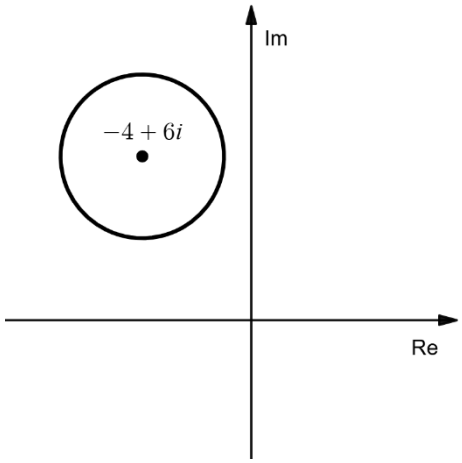
$32m-60m^2=0$

$m(8m-15)=0$

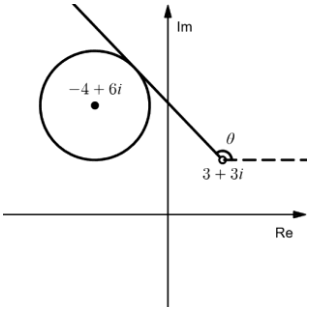
$\therefore m=0$ or $m=\frac{15}{8}$

$\tan\theta=\frac{8}{15}\therefore\theta=\arctan\left(\frac{8}{15}\right)$ as req.

36) a)



36) b)



Need the values of θ for which the line intersects with the circle exactly once.

$(x+4)^2+(y-6)^2=9$ and $y-3=m(x-3)$

$(x+4)^2+(mx-3m+3-6)^2=9$

$(x+4)^2+(mx-3m-3)^2=9$

$x^2+8x+16+m^2x^2-6m^2x+9m^2-6mx+18m+9=9$

$(1+m^2)x^2+(8-6m^2-6m)x+(9m^2+18m+16)=0$

Consider $b^2-4ac=0$:

$(8-6m^2-6m)^2-4(1+m^2)(9m^2+18m+16)=0$

$36m^4+72m^3-60m^2-96m+64-36m^4-72m^3-100m^2-72m-64=0$

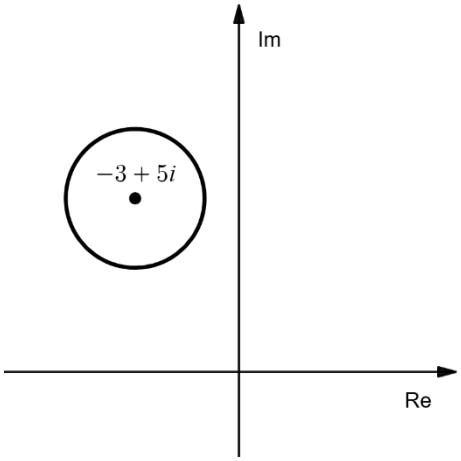
$-160m^2-168m=0$

$m(20m+21)=0$

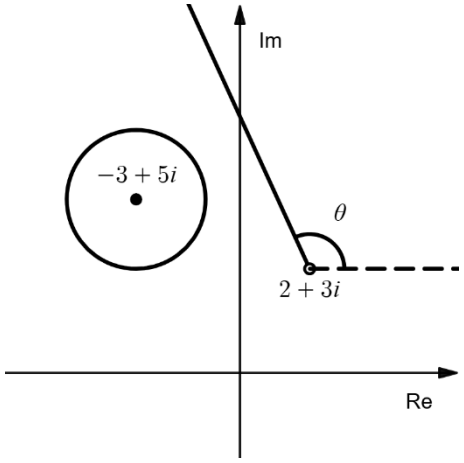
$\therefore m=0$ or $m=-\frac{21}{20}$

Which gives $\arctan\left(\frac{20}{21}\right)$ $\therefore \theta=\frac{\pi}{2}+\arctan\left(\frac{20}{21}\right)$ as req.

37) a)



37) b)



Need the values of θ for which the line intersects with the circle.

$(x+3)^2+(y-5)^2=4$ and $y-3=m(x-2) \Rightarrow y=mx-2m+3$

$(x+3)^2+(mx-2m+3-5)^2=4$

$(x+3)^2+(mx-2m-2)^2=4$

$x^2+6x+9+m^2x^2-4m^2x-4mx+4m^2+8m+4=4$

$(1+m^2)x^2+(6-4m-4m^2)x+(4m^2+8m+9)=0$

Consider $b^2-4ac=0$:

$(6-4m-4m^2)^2-4(1+m^2)(4m^2+8m+9)=0$

$16m^4+32m^3-32m^2-48m+36-16m^2-32m-36-16m^4-32m^3-36m^2=0$

$84m^2+80m=0$

$m(21m+20)=0$

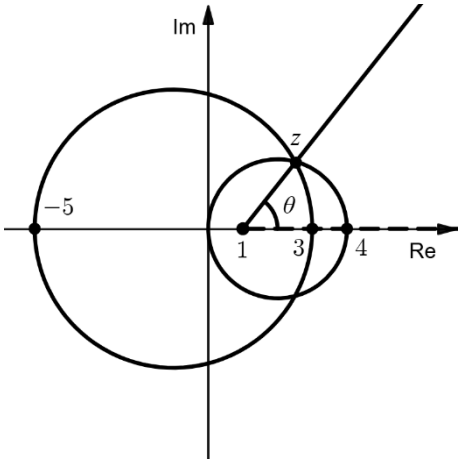
$\therefore m=0$ or $m=-\frac{20}{21}$ so $\tan\phi=0$ or $\tan\phi=-\frac{20}{21}$ where $\theta+\phi=\pi$

So $\theta=\pi-\tan\phi=\pi$ or $\theta=\pi-\arctan\left(\frac{20}{21}\right)=2.38$ or $\theta=\frac{\pi}{2}+\arctan\left(\frac{21}{20}\right)=2.38$

$\therefore \frac{\pi}{2}+\arctan\left(\frac{21}{20}\right)\leq\theta\leq\pi$ or $2.38\leq\theta\leq3.14$

38) a)

There are two values of z which satisfy the loci, one of which is shown in the diagram below:



Solve $(x-2)^2+y^2=4$ and $(x+1)^2+y^2=16$ simultaneously:

$(x-2)^2+16-(x+1)^2=4$

$x^2-4x+4+16-x^2-2x-1-4=0$

$-6x+15=0 \therefore x=\frac{5}{2}$ so $y=\pm\frac{\sqrt{15}}{2}$

$z=\frac{5}{2}+\frac{\sqrt{15}}{2}\mathrm{i}$ and $z=\frac{5}{2}-\frac{\sqrt{15}}{2}\mathrm{i}$

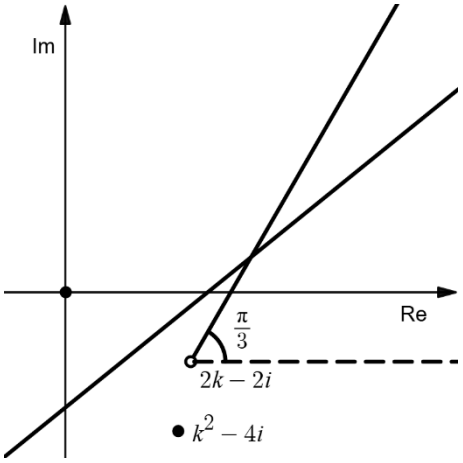
$\arctan\left(\frac{\frac{\sqrt{15}}{2}}{\frac{5}{2}-1}\right)=\arctan\left(\frac{\sqrt{15}}{3}\right)=0.912$ (3 SF)

$\therefore \theta=0.912$ or $\theta=-0.912$ (3 SF)

38) b)

$-0.912<\theta<0.912$

39) a)



Midpoint: $\left(\frac{1}{2}k^2, -2\right)$ and gradient: $m = -\frac{4}{k^2}$

$y + 2 = \frac{k^2}{4}\left(x - \frac{1}{2}k^2\right) \Rightarrow y = \frac{k^2}{4}x - \frac{k^4}{8} - 2$

Half-line:

$y + 2 = \tan\frac{\pi}{3}(x - 2k) \Rightarrow y = \sqrt{3}x - 2\sqrt{3}k - 2$

Solve $y = \frac{k^2}{4}x - \frac{k^4}{8} - 2$ and $y = \frac{k^2}{4}x - \frac{k^4}{8} - 2$ simultaneously:

$\frac{k^2}{4}x - \frac{k^4}{8} - 2 = \sqrt{3}x - 2\sqrt{3}k - 2$

$\frac{k^2}{4}x - \sqrt{3}x = -2\sqrt{3}k + \frac{k^4}{8}$

$\left(\frac{k^2}{4} - \sqrt{3}\right)x = -2\sqrt{3}k + \frac{k^4}{8}$

$\left(2k^2 - 8\sqrt{3}\right)x = -16\sqrt{3}k + k^4$

$x = \frac{k^4 - 16\sqrt{3}k}{2k^2 - 8\sqrt{3}}$ as req

39) b)

$\mathrm{Im}(z) = 0$ when $y = 0$

$0 = \sqrt{3}x - 2\sqrt{3}k - 2 \Rightarrow x = \frac{2\sqrt{3}}{3} + 2k$

and $0 = \frac{k^2}{4}x - \frac{k^4}{8} - 2 \Rightarrow 0 = \frac{k^2}{4}\left(\frac{2\sqrt{3}}{3} + 2k\right) - \frac{k^4}{8} - 2$

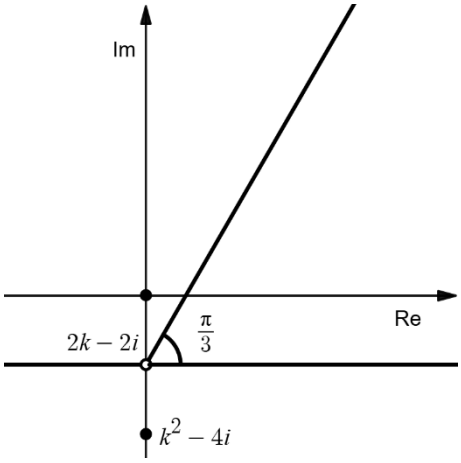
$0 = \frac{\sqrt{3}}{6}k^2 + \frac{1}{2}k^3 - \frac{k^4}{8} - 2$

Solving gives $k = 1.6118\ldots$ and $k = 4.3363\ldots$

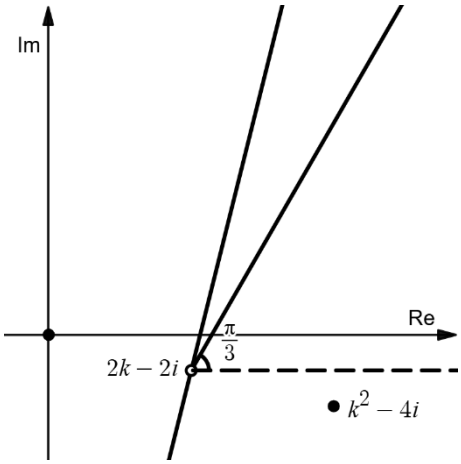
This gives $x = 4.38$ and $x = 9.83$

$\therefore \mathrm{Re}(z) = 4.38$ when $k = 1.61$ and $\mathrm{Re}(z) = 9.83$ when $k = 4.34$

39) c)



When $k = 0$, $\mathrm{Re}(z) = 0$ so the half-line starts at $-2\mathrm{i}$ which means this point is not included.



When $k = 4$, the midpoint between 0 and $k^2 - 4\mathrm{i}$ is $8 - 2\mathrm{i}$ which is the start point for the half-line which means this point is not included.

39) d)

The two lines are parallel when $\frac{k^2}{4} = \sqrt{3}$ so when $k = \pm 2 \times 3^{\frac{1}{4}}$

Consider k as it varies between $-2 \times 3^{\frac{1}{4}}$, 0, $2 \times 3^{\frac{1}{4}}$ and 4

$\therefore k < -2 \times 3^{\frac{1}{4}}$, $0 < k < 2 \times 3^{\frac{1}{4}}$, $k > 4$

40) a)

Let $z = x + y\text{i}$

$\frac{z-4}{z} = \frac{x+y\text{i}-4}{x+y\text{i}} \times \frac{x-y\text{i}}{x-y\text{i}} = \frac{x^2+xy\text{i}-4x-xy\text{i}+y^2+4y\text{i}}{x^2+y^2} = \frac{x^2-4x+y^2+4y\text{i}}{x^2+y^2}$

$\therefore \frac{z-4}{z} = \frac{x^2-4x+y^2}{x^2+y^2} + \frac{4y}{x^2+y^2}\text{i}$ as req.

40) b)

Since $\arg\left(\frac{z-4}{z}\right) = \frac{\pi}{3}$ then $\tan\left(\frac{\pi}{3}\right) = \frac{\frac{4y}{x^2+y^2}}{\frac{x^2-4x+y^2}{x^2+y^2}} = \frac{4y}{x^2-4x+y^2}$

So $\sqrt{3} = \frac{4y}{x^2-4x+y^2}$

$\sqrt{3}x^2 - 4\sqrt{3}x + \sqrt{3}y^2 = 4y$

$\sqrt{3}x^2 - 4\sqrt{3}x + \sqrt{3}y^2 - 4y = 0$

$\sqrt{3}(x-2)^2 - 4\sqrt{3} + \sqrt{3}\left(y - \frac{2\sqrt{3}}{3}\right)^2 - \frac{4\sqrt{3}}{3} = 0$

$(x-2)^2 - 4 + \left(y - \frac{2\sqrt{3}}{3}\right)^2 - \frac{4}{3} = 0$

$(x-2)^2 + \left(y - \frac{2\sqrt{3}}{3}\right)^2 = \frac{16}{3}$ which is an equation of a circle

The circle crosses the real axis at $z = 0$ and $z = 4$

Since the argument is positive, only z must lie above the real axis where $z = 0$ and $z = 4$ have no solutions to $\arg\left(\frac{z-4}{z}\right) = \frac{\pi}{3}$

It is the circle for which only non-zero positive imaginary numbers are included.

40) c)

